INSIGHTX | PROJECT REPORT

Drug Sales Demand Prediction

Exploratory data analysis and machine learning to forecast daily pharmacy demand and plan stock

Models: Linear Regression | Random Forest Regressor

Tools: Python | pandas | scikit-learn | matplotlib | seaborn

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1. Executive summary

A pharmacy that orders too little stock loses sales; one that orders too much ties up money and risks expired medicines. This project uses a year of daily sales records to predict **how many units of a drug will be sold on a given day**, so that stock can be planned in advance.

365	16,057	0.51	6.7
days of sales data	units sold in total	R-squared of the best model (Linear Regression)	average error in units per day (MAE)

What was done. The sales data was explored, calendar features were created, and two models, **Linear Regression** and a tuned **Random Forest**, were trained on the first 292 days and tested on the last 73 days (2023-10-20 to 2023-12-31). A naive "always predict the average" model serves as a yardstick.

Main findings

- **Linear Regression is the best model:** on the unseen test period it is off by 6.7 units per day on average (MAPE 13%), about 42% better than the naive baseline (11.6 units). R-squared is 0.51, against -0.32 for the baseline.
- **Price is the strongest driver of demand.** Cheap drugs (under 50) sell 53 units per day on average, expensive ones (over 110) only 35.
- **Seasonality:** Antibiotics and Antivirals sell 57 units per day in winter (Dec to Feb) but 30 in summer (Jun to Aug).
- **Weekends are quieter:** 38 units per day on weekends versus 46 on weekdays.
- The model is turned into a **demand-planning table** (predicted daily demand per category and month) that supports ordering decisions.

Important: the dataset is **synthetic** (see Section 3). The results demonstrate the workflow and the kind of insights it can deliver; they should not be quoted as real-world forecasting accuracy.

2. Introduction

2.1 Problem statement

Demand for medicines changes with the season, the day of the week, the price and the type of drug. Pharmacy managers who rely on gut feeling tend to over-stock or run out of popular items. The task is to predict the daily quantity sold from information that is known in advance: the date, the drug, its category and its price.

2.2 Objectives

- Explore the sales data and find the patterns that influence demand.
- Build and compare regression models that predict the daily quantity sold.
- Measure how much better the models are than a naive guess, using a realistic time-based test.
- Identify the most important demand drivers and explain them in plain language.
- Turn the model into a demand-planning table that helps decide how much stock to order.

2.3 Why these two models?

Linear Regression is a simple and transparent baseline: every factor gets a coefficient that states by how many units it raises or lowers daily sales. **Random Forest Regressor** averages hundreds of decision trees and can capture non-linear patterns and interactions, for example "winter and Antibiotics". Comparing them shows whether the extra complexity is worth it.

3. Dataset

The file `data.csv` holds **365 daily records** (1 January to 31 December 2023) with five columns.

Column	Description
Date	Day of the sale (one record per day)
Drug_ID	Drug sold: Drug_A, Drug_B, Drug_C or Drug_D
Category	Antibiotics, Analgesics, Antivirals or Cardiology
Quantity_Sold	**Target:** units sold that day
Unit_Price	Selling price per unit (about 10 to 150)

3.1 Nature of the data

The dataset is **synthetic**. Dates, drugs, categories and prices are generated as in the original project notebook. In that original version the quantity was also purely random, which means it had no relationship to date, drug or price; tested on the same time-based split, such data gives an R-squared of about -0.01 for Linear Regression and -0.18 for the Random Forest, i.e. no model can learn anything. For this reason **Quantity_Sold now follows realistic patterns plus random noise**: popularity differences between drugs and categories, a winter peak for Antibiotics and Antivirals, lower weekend sales and lower demand at higher prices. The data contains no real sales figures, so the scores should not be read as real-world accuracy.

3.2 Data quality

The data has no missing values and no duplicate rows, and every day appears exactly once. Daily quantity averages 44.0 units (range 5 to 78).

4. Methodology

Sales form a time series, so the data is split **by time, not at random**: the models learn from the first 80% of the year and are tested on the last 20%, exactly the situation in practice, where the future is predicted from the past. All preprocessing is bundled with each model in one scikit-learn *Pipeline*.

Step	What was done
1. Exploration	Distributions, averages by category and drug, monthly seasonality, weekday and price effects, correlations, revenue.
2. Feature engineering	From Date: **Month, Quarter, DayOfWeek, Is_Weekend**, and the cyclical **Month_Sin / Month_Cos** (so that December and January are neighbours).
3. Preprocessing	Drug_ID and Category: one-hot encoding (label encoding would imply a false order). Unit_Price: MinMax scaling to 0-1, fitted on training data only.
4. Split	Chronological: 292 training days (2023-01-01 to 2023-10-19) and 73 test days (2023-10-20 to 2023-12-31).
5. Modelling	Naive baseline (training mean), Linear Regression, and Random Forest with grid search using 5-fold **time-series cross-validation**.
6. Evaluation	MAE, RMSE, R-squared and MAPE on the test period; cross-validated MAE on the training period.
7. Interpretation	Permutation importance, Linear Regression coefficients, and a predicted demand table by category and month.

4.1 Random Forest tuning

Twelve combinations of *n_estimators* (100, 300), *max_depth* (4, 8, unlimited) and *min_samples_leaf* (1, 3) were compared. The best setting was **n_estimators = 300, max_depth = 8, min_samples_leaf = 1**, with a cross-validated MAE of 8.26 units.

4.2 How performance is measured

Metric	Meaning for this project
MAE	Average error in units per day. Easy to interpret: "the forecast is typically off by this many units".
RMSE	Like MAE, but large misses are punished more.
R-squared	Share of the variation in daily sales explained by the model (0 = no better than the average, 1 = perfect; negative = worse than the average).
MAPE	Average error as a percentage of the real quantity.

5. Exploratory data analysis

5.1 Distributions

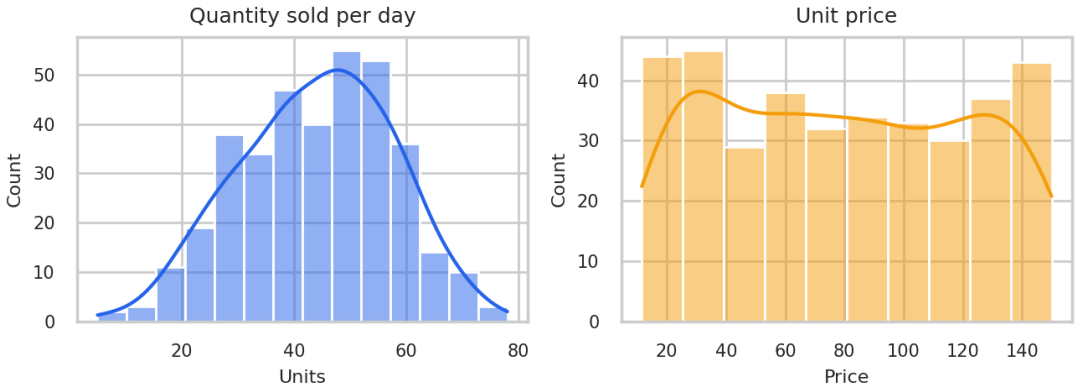


Figure 1. Distribution of daily quantity sold and of unit price.

Daily quantity is roughly bell-shaped around 44 units. Unit price is spread evenly between about 10 and 150, so both cheap and expensive sales are well represented.

5.2 Demand by category and by drug

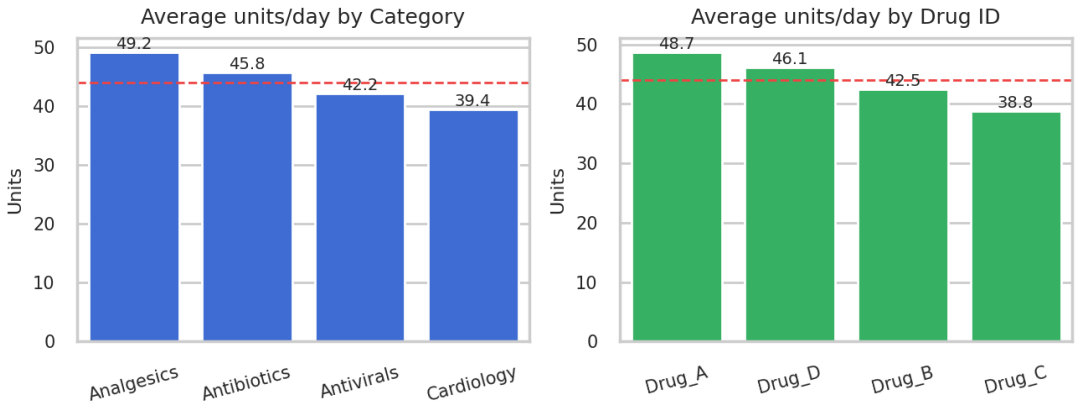


Figure 2. Average units sold per day by category and by drug (red line = overall average).

Analgesics is the best-selling category (49.2 units per day) and Cardiology the weakest (39.4). Among drugs, Drug_A leads (48.7) and Drug_C trails (38.8). In revenue terms (quantity times price) the order is: Analgesics (333K), Cardiology (320K), Antibiotics (293K), Antivirals (207K).

5.3 Seasonality

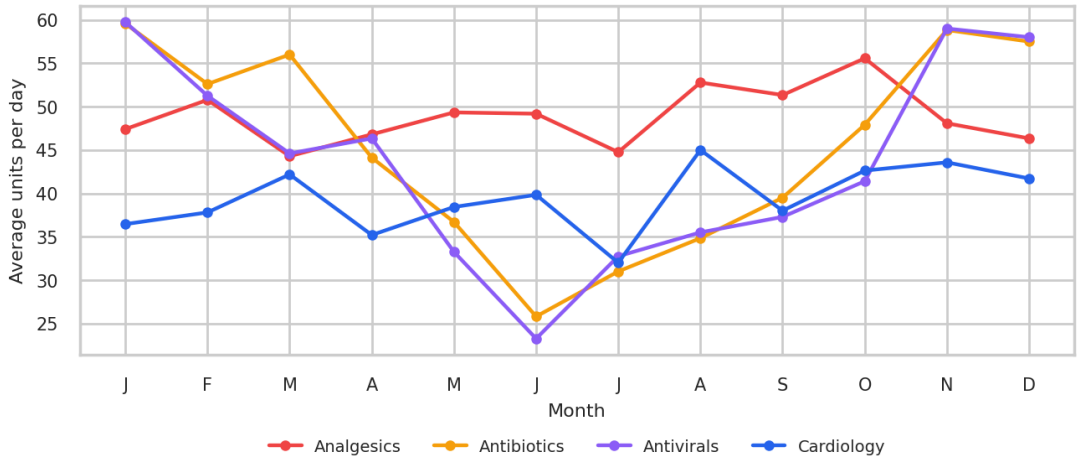


Figure 3. Average daily quantity sold per month and category.

Antibiotics and Antivirals follow a clear yearly cycle: high in winter (57 units per day in Dec to Feb) and low in mid-year (30 in Jun to Aug). Analgesics and Cardiology show no clear seasonal pattern; their month-to-month changes look like random fluctuation.

5.4 Weekday and price effects

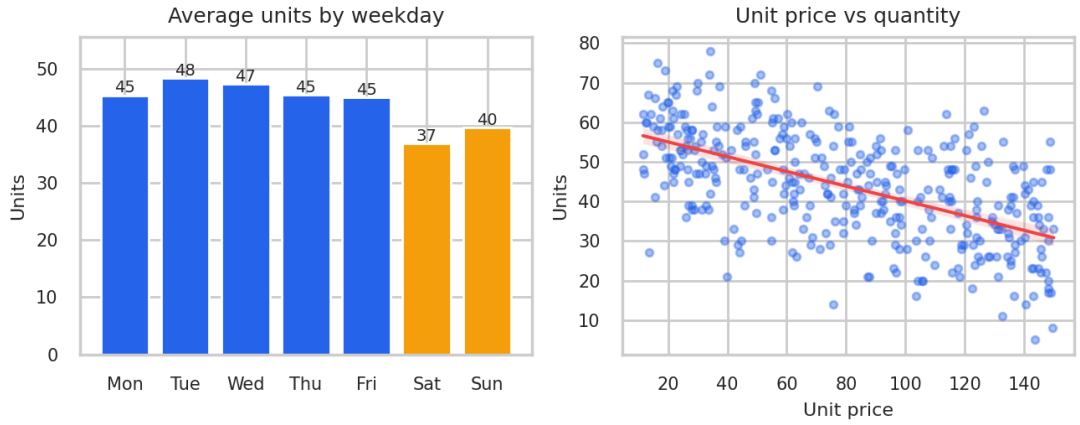


Figure 4. Average quantity by weekday (weekend in amber) and the relationship between price and quantity.

- **Weekends:** 38 units per day on average, compared with 46 on weekdays.
- **Price:** the correlation between unit price and quantity is -0.58 (clearly negative). Drugs priced under 50 sold 53 units per day, those over 110 only 35.

5.5 Correlations

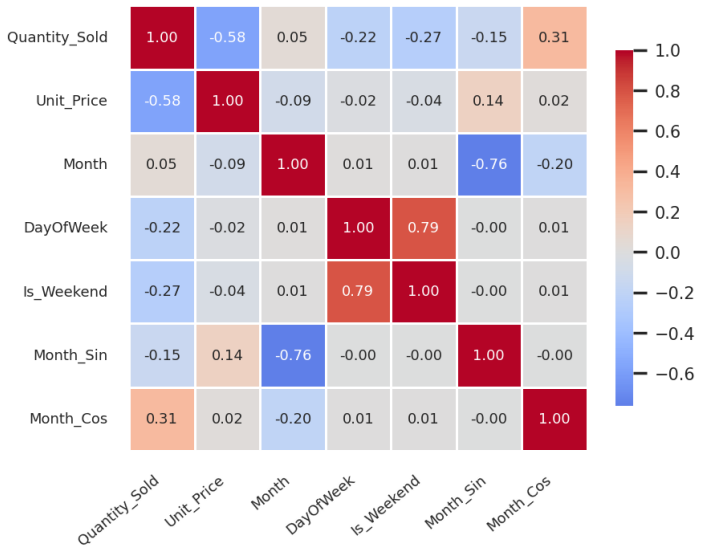


Figure 5. Correlation matrix of the numeric columns.

Price has the strongest link to quantity (-0.58, negative). Month_Cos (+0.31, the winter peak) and the weekend flag (-0.27) follow. The seasonal link is only moderate in this overall view because the winter swing affects just two of the four categories; the models can use it in combination with the category.

6. Model results

All models were evaluated on the 73 days at the end of the year that were never used for training.

Model	MAE	RMSE	R-squared	MAPE (%)	Train R-sq.	CV MAE
Naive baseline (mean)	11.60	13.85	-0.319	23.9	0.000	11.11
Linear Regression	**6.68**	**8.48**	**0.505**	**13.4**	**0.697**	**7.46**
Random Forest (tuned)	7.67	9.70	0.354	15.3	0.933	8.26

MAE, RMSE and CV MAE are in units per day. The best model is in bold.

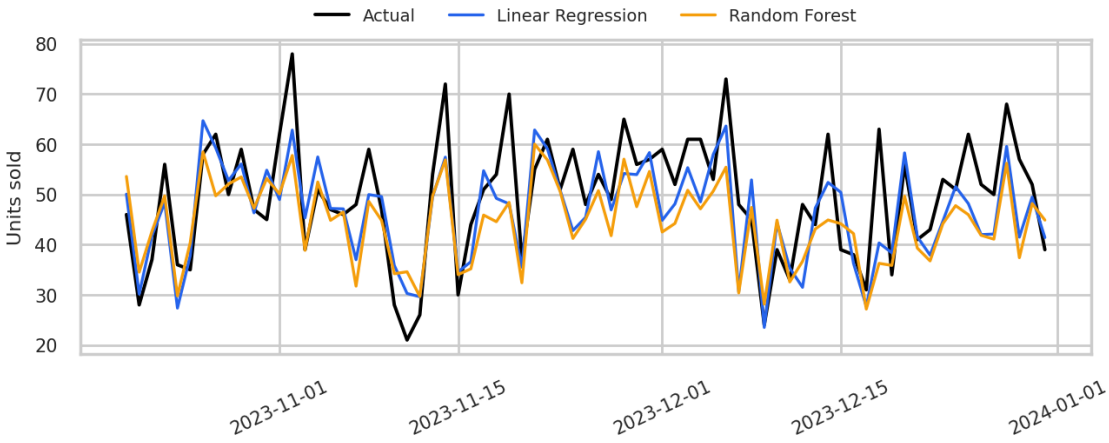


Figure 6. Actual and predicted daily quantity during the test period.

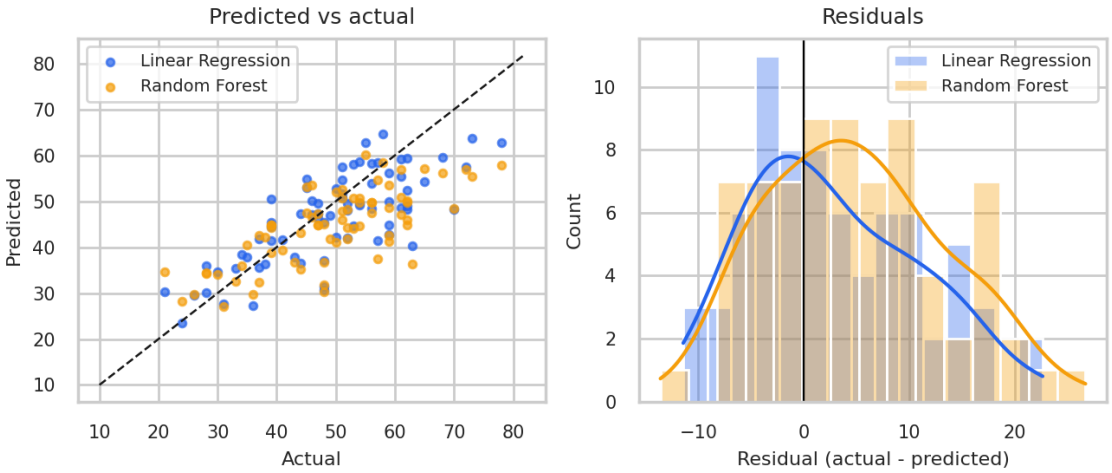


Figure 7. Predicted versus actual quantity (left) and distribution of the prediction errors (right).

6.1 Interpretation

- **Both models beat the naive baseline.** The baseline has an R-squared of -0.32 (worse than the test-period average) and a MAE of 11.6 units. The negative value suggests that the final months of the year differ from the average of the earlier months, i.e. seasonality matters.
- **Linear Regression is best** (MAE 6.68, R-squared 0.505). The cross-validated MAE points the same way (7.46 for Linear Regression versus 8.26 for the Random Forest).
- **Overfitting of the Random Forest.** It explains 93% of the variation on training days but only 35% on the test days. Linear Regression shows a smaller gap (70% versus 50%). With only 292 training days and effects that are mostly additive (price, weekend, drug and category offsets), the simpler model generalises better. More data would likely help the forest.
- **Typical accuracy:** Linear Regression is about 13% off on an average day. The test period has only 73 days, so the metrics themselves are somewhat noisy.

7. What drives demand?

Permutation importance shuffles one original feature at a time and measures how much the Random Forest's R-squared drops: the bigger the drop, the more the model depends on that feature.

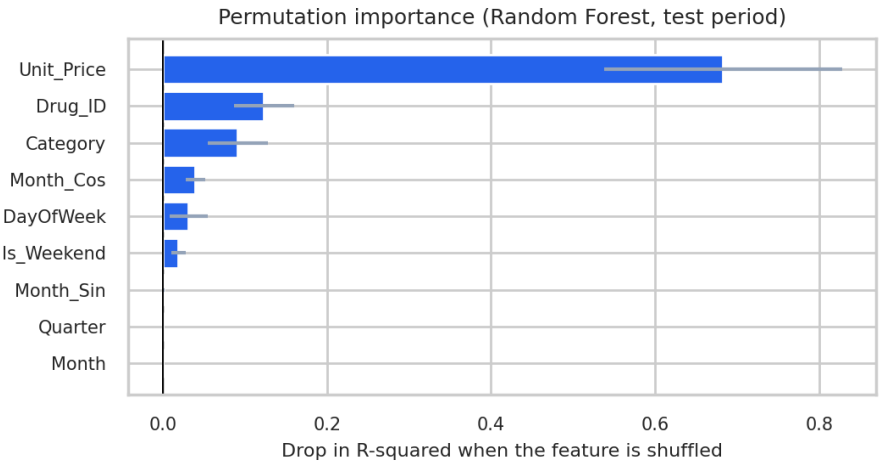


Figure 8. Permutation importance of the original features (test period; bars show the variation over 30 repeats).

Unit Price dominates (R-squared falls by 0.68 when it is shuffled), followed by **Drug ID** (0.12) and **Category** (0.09). The calendar features look small individually because several of them (Month, Quarter, Month_Sin, Month_Cos) describe the same thing, so their importance is shared between them.

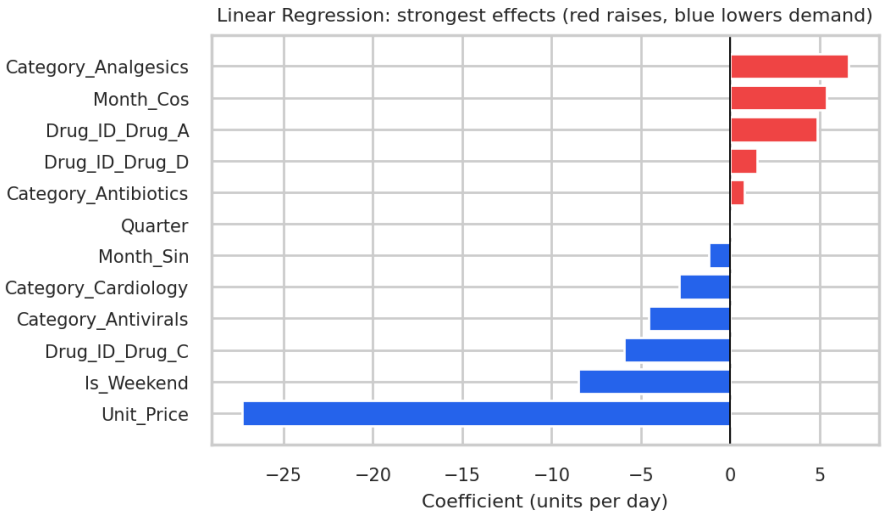


Figure 9. Linear Regression coefficients: change in daily units (red raises, blue lowers demand).

- **Price:** moving from the cheapest to the most expensive price lowers demand by about 27 units per day.
- **Weekend:** about 8.5 units fewer per day than on weekdays.
- **Drugs and categories:** Drug_A sells about 5 units more than average and Drug_C about 6 fewer; Analgesics sell about 7 more and Antivirals about 5 fewer than average.
- **Season:** the positive Month_Cos coefficient reflects the winter peak; it is an average over all categories, so the seasonal effect on Antibiotics and Antivirals alone is larger.

8. Demand planning

To turn the model into a tool, the Random Forest predicts the **average daily demand** for every category in every month (weekday sale, median price, averaged over the four drugs). A manager can read from this table how much stock each category needs in each season.

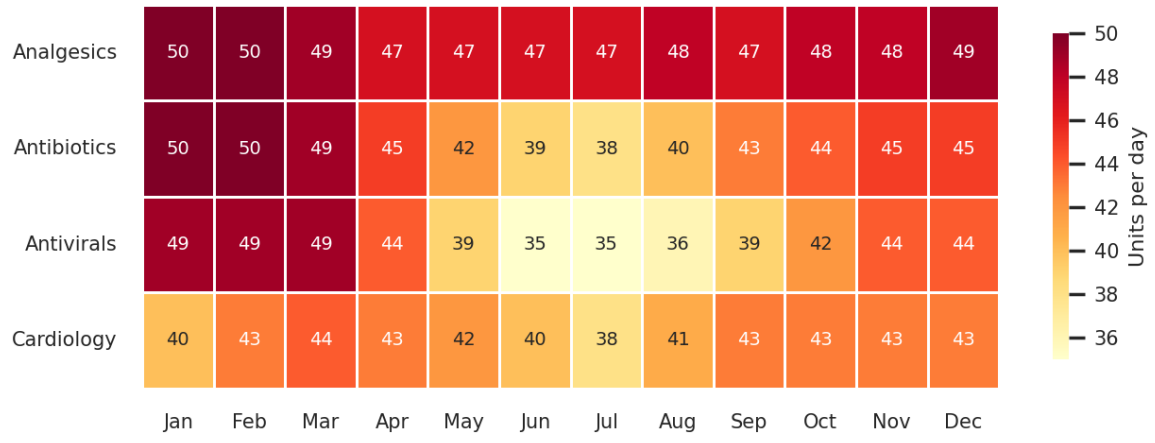


Figure 10. Predicted average daily demand (units) by category and month.

- **Analgesics:** highest in Jan (50 units per day), lowest in Apr (47).

- **Antibiotics:** highest in Jan (50 units per day), lowest in Jul (38).
- **Antivirals:** highest in Jan (49 units per day), lowest in Jun (35).
- **Cardiology:** highest in Mar (44 units per day), lowest in Jul (38).

8.1 Suggested actions

The following are proposals based on the model, to be adapted by the pharmacy; they are not outputs of the model itself.

- **Build up stock for Antibiotics and Antivirals before winter** (from about October) and reduce orders in mid-year.
- **Keep Analgesics and Cardiology stock steady** through the year; their predicted demand changes little.
- **Plan lighter replenishment around weekends**, since weekend demand is lower.
- **Treat price changes with care:** demand falls as price rises, so a discount can lift volume, but the effect on revenue (quantity times price) should be checked before acting.
- **Retrain regularly** (for example monthly) with new sales data so that forecasts follow changing patterns.

9. Limitations and future work

9.1 Limitations

- **Synthetic data.** The patterns were built in, so results show the workflow, not real-world accuracy.
- **Only one year.** The yearly cycle cannot be validated across several years, and the test period covers just the last 73 days.
- **Simplified record structure.** Each day has one sale record, and Drug_ID and Category are assigned independently. Real data would have one row per drug per day, with each drug belonging to one category.
- **Missing real-world factors.** Holidays, promotions, stock-outs, epidemics and weather are not included.
- **Price as a cause.** In real data, price changes usually come with other changes, so the price effect found here should not be taken as a guaranteed result of a price cut.

9.2 Future work

- Use several years of real sales data with one row per drug per day.
- Add holiday, promotion, weather and disease-outbreak features, and lagged sales (yesterday, last week).
- Try gradient boosting and dedicated time-series models such as SARIMA or Prophet.
- Produce forecast intervals (a range, not a single number) to set safety-stock levels.
- Build a small dashboard so that pharmacy staff can view forecasts without code.

10. Conclusion

This project shows end to end how daily sales records can be turned into a demand forecast. After exploring the data and engineering calendar features, two models were trained and tested on a realistic time-based split. **Linear Regression** performed best, with an average error of 6.7 units per day, about 42% better than a naive guess. Price, the type of drug and the category are the strongest demand drivers, with additional seasonal and weekend effects. A demand-planning table translates the predictions into practical stocking guidance. The next step is to repeat the analysis on several years of real sales data.

Appendix A. Repository contents

File	Description
README.md	Project overview, how to run, how to use the saved models.
data.csv	Synthetic daily sales dataset (365 rows, 5 columns).
Drug_Sales_Demand_Prediction.ipynb	Complete notebook: EDA, feature engineering, training, evaluation, importance, demand planning, model saving.
linear_regression_model.pkl	Trained Linear Regression pipeline (preprocessing + model).
random_forest_model.pkl	Trained and tuned Random Forest pipeline (preprocessing + model).
InsightX_Project_Report.pdf	This report.

How to reproduce

- Open `Drug_Sales_Demand_Prediction.ipynb` in Google Colab or Jupyter.
- Keep `data.csv` in the same folder (if it is missing, the notebook regenerates it).
- Run all cells. The trained models are written to the two .pkl files.

The models were created with scikit-learn 1.8.0. If a different version raises an error when loading a .pkl file, simply re-run the notebook to regenerate it.

Appendix B. References

- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5-32.
- Hyndman, R. J. and Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts.